



Oral Remdesivir delivery

Scientists are finding a way for Remdesivir to be delivered to the body as dry powder instead of injections. <http://digit.in/sep20-59>

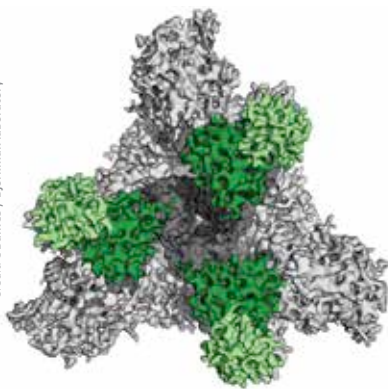


Ventilator alternative

High flow nasal canulas are great alternatives to ventilators, but require more PPE kits. <http://digit.in/sep20-60>

severe symptoms. When the virus initially infects a patient, the white blood cells use a sticky substance similar to spider silk, called Neutrophil Extracellular Traps (NETs) to capture the viruses. If the immune system overreacts and deploys too many white blood cells, the accumulation of NETs in the bloodstream can lead to clogging and inflammation of the tissues. The team examined the plasma of patients with severe symptoms, and found a correlation of

Credit: CBeames / Bjorkman laboratory



A SARS-CoV-2 trimer captured by a neutralising antibody

increased NETs in the bloodstream as well. Low blood oxygen levels, an independent marker of the severity of the disease, also tracked with higher quantities of NETs. In healthy or asymptomatic individuals, the NET activity was measurably much lower. A protein found in the umbilical cord of newborn babies, called the neonatal NET Inhibitory Factor (nNIF) can suppress the white blood cells from producing too many NETs. This protein could eventually become the basis for a treatment for the more severe cases.

HOW STEM CELLS CAN HELP

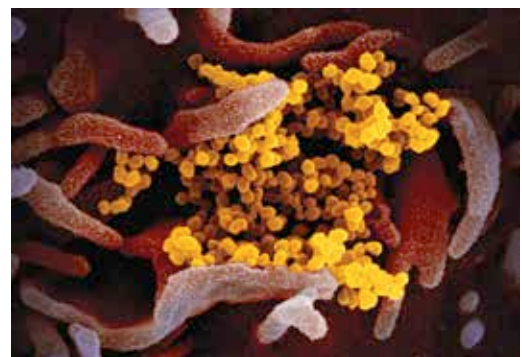
The patients with severe symptoms are known to have a severe immune response as well. A large number of cytokines are released in the body, which are used for signalling and regulation purposes of the immune system. A rush of these cytokines within the bloodstream is known as a cytokine storm, and causes an

inflammatory response in the body, which can damage organs and tissues. It can also lead to organ failure, most typically in the lungs. The cytokine storm can also cause patients to go into shock, and lead to fluid buildup in the lungs. This is why the patients with severe symptoms need to be put on ventilators. The cytokine storm is one of the main reasons that leads to death in those suffering from COVID-19.

Within the bone marrow of all humans, are the mesenchymal stem cells (MSCs), which are responsible for the repair of the bone tissue and cartilage. The MSCs are also present in the umbilical cords of newborns. In a study conducted in China, seven COVID-19 patients were injected with MSCs into the blood streams, and these patients fared better than those who did not receive the treatment. While the sample size was small, the treatment showed promise, which led to other similar studies around the world. A similar treatment was provided to some patients in Italy as well. Researchers from Texas A&M University and The University of Minnesota are conducting small scale trials of MSC treatments on some patients with severe symptoms in a few hospitals in the US. The treatment cannot be provided in the early stages of the infection, as it does not make sense to stifle the immune system from its natural reaction. It is only required on the patients with the worst symptoms. If the trials are successful, this promising treatment may be provided to more people.

Apart from being directly injected into the body, stem cells can also help in rapidly testing the effects of different kinds of drugs. To do this, researchers use what are called organoids, grown from stem cells that are miniature lungs that have the same kind of cells and architecture as human lungs, but on a smaller scale.

These organoids can allow researchers to find out how the virus infects and affects the cells of an actual person, and then used to screen a large number of drugs for potential treatment options. One of the institutions conducting the screening for drugs using lung organoids, is the University of California, Los Angeles (UCLA). Other teams at UCLA are using stem cells in various ways. This includes studying the immune response and the cytokine storm that leads to the more severe symptoms. The idea is to use the stem cells that create blood in the body, which then respond to the SARS-CoV-2 virus by producing dendritic cells, which then shred the virus up and present the debris to another specialised type of cell called T-cells. The T-cells produce the immune response, and the researchers are looking at which specific structures on the virus are responsible for triggering the cytokine storm. The T-cells remember the viruses that they have encountered before, and can allow the immune system to react appropriately when the virus is encountered again. In the older patients, the body's



A scanning electron microscope image of the SARS-CoV-2 virus (the yellow globules)

Credit: National Institute of Allergy and Infectious Diseases

ability to create T-cells reduces, and the protection offered by them does not last for very long. Which is why, researchers are developing molecules that promote the production of T-cells within the body, which can be added to any vaccine developed in the future, that will help even the older patients develop long lasting protection against SARS-CoV-2.